Estimated Annual Agricultural Pesticide Use
Bingham County, Idaho 2014-2018

24 July 2026

Summary

Bingham County sits at the heart of Idaho's Snake River Plain and is the state's leading potato-producing county — Idaho itself grows roughly a third of the United States' potato crop, and the "Famous Idaho Potato" identity is centered on counties like Bingham. The county also produces substantial wheat, barley, and sugar beets under irrigation, playing a role in Idaho agriculture comparable to Yakima County's for Washington's tree-fruit and hop industries in the companion Washington chapter of this repository. Bingham is also Idaho's highest-use county by total estimated pesticide mass applied of any window in this repository's county-level data, making it an unambiguous choice on both agronomic-identity and raw-use grounds.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Bingham County, Idaho** during 2018, as reported by USGS Estimated Annual Agricultural Pesticide Use data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by Thelin and Stone (2013) and Baker and Stone (2015):

> "EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD." *-Pesticide National Synthesis Project webpage*

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. "detailed interpretation of where and how much use occurs within a county is not appropriate").
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial

Table 1: Chemical class abbreviation key (used in the Class column of Table 2) *(continued)*

Code	Chemical class
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound
OTH	Other
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Bingham County, ID, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	METAM POTASSIUM	DTC	1608250	732835	517983
2	METAM	DTC	750850	1096486	1191112
3	SULFURIC ACID	INO	312573	578007	664520
4	CHLOROPICRIN	OC	306891	135441	101582
5	DICHLOROPROPENE	OC	165580	257701	238598
6	GLYPHOSATE	OTH	45130	36390	40192
7	2,4-D	CPX	37048	17862	13830
8	CHLOROTHALONIL	OTH	22724	20962	19081
9	PENDIMETHALIN	OTH	19518	19116	17527
10	METRIBUZIN	TRZ	16246	16634	15539
11	EPTC	TC	13454	17373	18580
12	MCPA	CPX	12388	10052	12441
13	MALEIC HYDRAZIDE	OTH	10821	7060	8254
14	DIMETHENAMID & DIMETHENAMID-P	OTH	10747	10624	9466
15	DIMETHENAMID-P	OTH	10747	10181	9200
16	MANCOZEB	DTC	10516	9210	16353
17	PHOSPHORIC ACID	INO	8507	8173	14143
18	METHOMYL	CB	7913	6607	5187
19	METOLACHLOR & METOLACHLOR-S	OTH	7685	6052	5602
20	METOLACHLOR-S	OTH	6969	5580	5005
21	BROMOXYNIL	OTH	5785	3955	4329
22	CHLORPYRIFOS	OP	5663	5280	6136
23	FLUROXYPYR	OTH	5078	5053	4711
24	QUINTOZENE	OTH	4288	2384	2384
25	PYRIMETHANIL	OTH	3994	3375	3356
26	DIQUAT	DIP	3561	3375	4271
27	SULFUR	INO	3128	8189	8175
28	PROPARGITE	OTH	3031	3031	4692
29	AZOXYSTROBIN	OTH	2940	2570	2702
30	GLUFOSINATE	OTH	2911	1466	1163
31	BOSCALID	OTH	2828	2962	3171
32	IMIDACLOPRID	OTH	2369	2330	2805
33	PYRACLOSTROBIN	OTH	2121	1665	1879
34	FLUOPYRAM	OTH	1954	1588	1338
35	ACETOCHLOR	OTH	1757	745	837
36	PARAQUAT	DIP	1647	3101	4122
37	OXAMYL	CB	1554	1460	5591
38	PINOXADEN	OTH	1514	1213	1074
39	TRIFLURALIN	OTH	1448	1469	1088
40	ATRAZINE	TRZ	1272	1809	1117
41	PICLORAM	OTH	1268	442	272
42	PROPICONAZOLE	OTH	1268	1292	1315

Table 2: Top 80 EPest-high Estimates, Bingham County, ID, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) *(continued)*

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
43	PETROLEUM OIL	OTH	1247	74016	44812
44	FLUTOLANIL	OTH	965	2427	2462
45	CLOPYRALID	OTH	955	1320	1181
46	HEXAZINONE	TRZ	886	890	589
47	THIOPHANATE-METHYL	OTH	805	1515	1136
48	DIFENOCONAZOLE	OTH	763	589	862
49	FLUXAPYROXAD	OTH	759	534	448
50	DIURON	OTH	742	849	743
51	DIMETHOATE	OP	735	4211	4134
52	MEFENOXAM	OTH	719	1033	986
53	METOLACHLOR	OTH	716	472	596
54	CARFENTRAZONE-ETHYL	OTH	703	493	333
55	CYFLUTHRIN	PYR	696	341	218
56	DICAMBA	OTH	686	709	618
57	METIRAM	DTC	671	671	1234
58	CYPROCONAZOLE	OTH	535	342	342
59	FLUAZINAM	OTH	524	521	914
60	PROTHIOCONAZOLE	OTH	522	257	173
61	TEBUCONAZOLE	OTH	467	661	417
62	CALCIUM POLYSULFIDE	INO	442	210	156
63	METALAXYL	OTH	440	440	1034
64	MANDIPROPAMID	OTH	405	320	425
65	THIFENSULFURON	OTH	354	316	260
66	TEBUTHIURON	OTH	351	351	351
67	SPIROTETRAMAT	OTH	349	289	177
68	PYROXASULFONE	OTH	326	183	540
69	CYHALOTHRIN-LAMBDA	PYR	321	459	354
70	METCONAZOLE	OTH	311	392	312
71	PYRASULFOTOLE	OTH	309	145	195
72	COPPER HYDROXIDE	INO	307	1304	1210
73	ZETA-CYPERMETHRIN	PYR	292	255	289
74	CLETHODIM	OTH	276	142	203
75	PICOXYSTROBIN	OTH	250	198	195
76	LINURON	OTH	234	247	153
77	FLUCARBAZONE	OTH	198	113	71
78	ALPHA CYPERMETHRIN	PYR	193	230	154
79	HEXYTHIAZOX	OTH	193	226	189
80	COPPER OXYCHLORIDE	INO	175	395	395

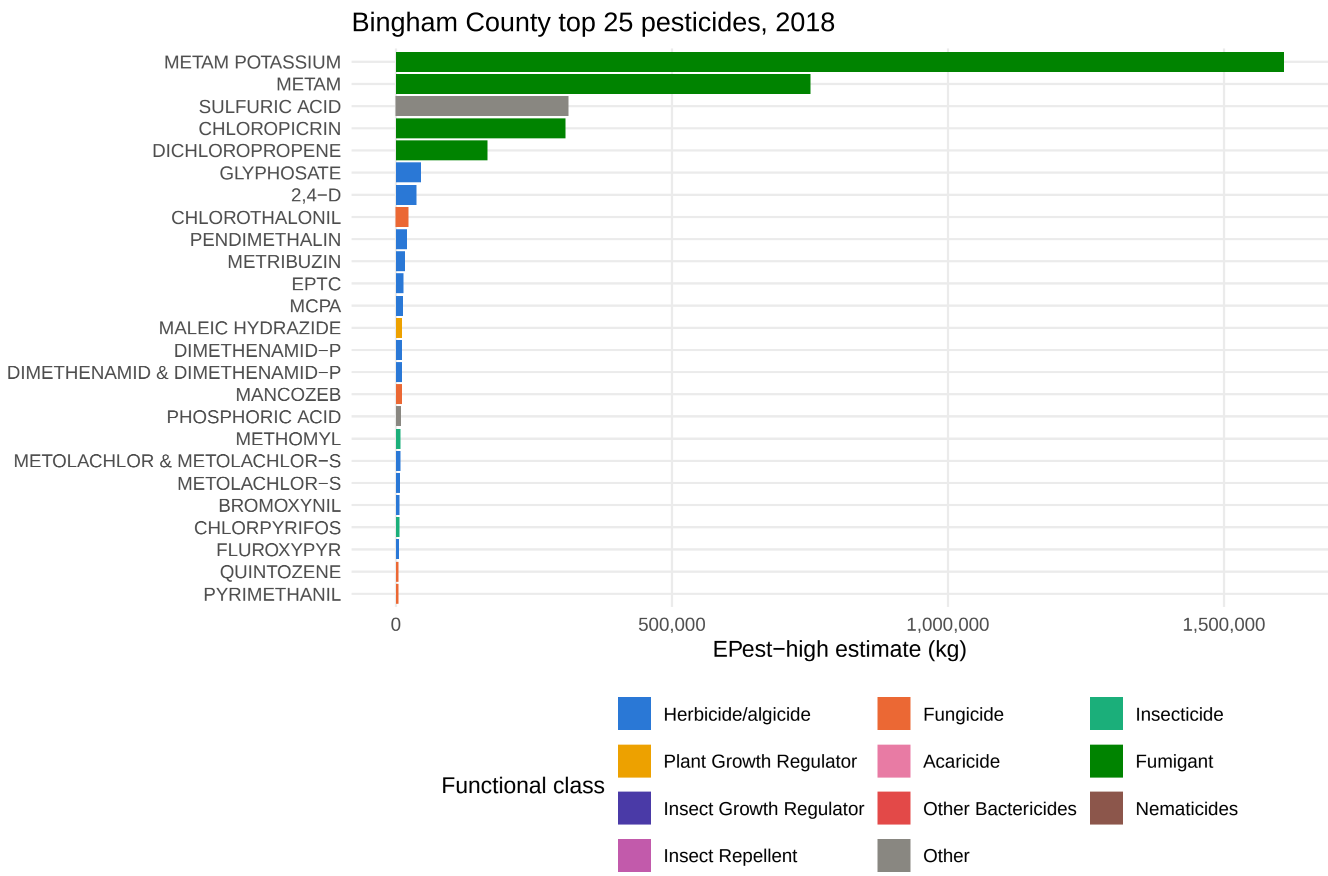

Figure 1: Top 25 compounds in Bingham County by EPest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Bingham County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM POTASSIUM	1608250	1608250
2	METAM	750850	750850
3	SULFURIC ACID	312573	312573
4	CHLOROPICRIN	43	306891
5	DICHLOROPROPENE	165580	165580
6	GLYPHOSATE	28734	45130
7	2,4-D	36961	37048
8	CHLOROTHALONIL	22724	22724
9	PENDIMETHALIN	17737	19518
10	METRIBUZIN	16170	16246
11	EPTC	13454	13454
12	MCPA	12388	12388
13	MALEIC HYDRAZIDE	0	10821
14	DIMETHENAMID & DIMETHENAMID-P	8631	10747
15	DIMETHENAMID-P	8631	10747
16	MANCOZEB	10516	10516
17	PHOSPHORIC ACID	8507	8507
18	METHOMYL	1	7913
19	METOLACHLOR & METOLACHLOR-S	7207	7685
20	METOLACHLOR-S	6562	6969
21	BROMOXYNIL	5785	5785
22	CHLORPYRIFOS	5642	5663
23	FLUROXYPYR	5062	5078
24	QUINTOZENE	0	4288
25	PYRIMETHANIL	3994	3994
26	DIQUAT	3561	3561
27	SULFUR	3125	3128
28	PROPARGITE	0	3031
29	AZOXYSTROBIN	2940	2940
30	GLUFOSINATE	2911	2911
31	BOSCALID	2828	2828
32	IMIDACLOPRID	2368	2369
33	PYRACLOSTROBIN	2112	2121
34	FLUOPYRAM	1899	1954
35	ACETOCHLOR	0	1757
36	PARAQUAT	38	1647
37	OXAMYL	1554	1554
38	PINOXADEN	1514	1514
39	TRIFLURALIN	75	1448
40	ATRAZINE	1	1272
41	PICLORAM	1205	1268
42	PROPICONAZOLE	1268	1268
43	PETROLEUM OIL	1247	1247
44	FLUTOLANIL	965	965
45	CLOPYRALID	379	955
46	HEXAZINONE	0	886
47	THIOPHANATE-METHYL	64	805
48	DIFENOCONAZOLE	492	763
49	FLUXAPYROXAD	755	759
50	DIURON	0	742

Table 4: Top 50 Bingham County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	1096486	1096486
2	METAM POTASSIUM	732835	732835
3	SULFURIC ACID	578007	578007
4	DICHLOROPROPENE	257701	257701
5	CHLOROPICRIN	16170	135441
6	PETROLEUM OIL	1865	74016
7	ETHOPROPHOS	0	50129
8	PROPAMOCARB HCL	0	44396
9	GLYPHOSATE	30886	36390
10	CHLOROTHALONIL	20962	20962
11	PENDIMETHALIN	16456	19116
12	2,4-D	16971	17862
13	EPTC	17225	17373
14	METRIBUZIN	16567	16634
15	DIMETHENAMID & DIMETHENAMID-P	9096	10624
16	DIMETHENAMID-P	9096	10181
17	MCPA	10008	10052
18	MANCOZEB	9210	9210
19	SULFUR	1158	8189
20	PHOSPHORIC ACID	8173	8173
21	MALEIC HYDRAZIDE	0	7060
22	METHOMYL	1	6607
23	METOLACHLOR & METOLACHLOR-S	5490	6052
24	METOLACHLOR-S	5254	5580
25	CHLORPYRIFOS	5098	5280
26	FLUROXYPYR	5034	5053
27	DIMETHOATE	893	4211
28	BROMOXYNIL	3914	3955
29	PYRIMETHANIL	3375	3375
30	DIQUAT	3375	3375
31	PARAQUAT	39	3101
32	PROPARGITE	0	3031
33	BOSCALID	2962	2962
34	AZOXYSTROBIN	2570	2570
35	FLUTOLANIL	2427	2427
36	QUINTOZENE	240	2384
37	IMIDACLOPRID	2330	2330
38	ATRAZINE	194	1809
39	SPIROMESIFEN	0	1678
40	PYRACLOSTROBIN	1657	1665
41	NALED	0	1650
42	FLUOPYRAM	1569	1588
43	THIOPHANATE-METHYL	21	1515
44	TRIFLURALIN	181	1469
45	GLUFOSINATE	1466	1466
46	OXAMYL	1460	1460
47	DIMETHENAMID	0	1329
48	CYAZOFAMID	0	1328
49	CLOPYRALID	177	1320
50	COPPER HYDROXIDE	802	1304

Table 5: Top 50 Bingham County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	1191110	1191112
2	SULFURIC ACID	664520	664520
3	METAM POTASSIUM	517983	517983
4	DICHLOROPROPENE	238598	238598
5	CHLOROPICRIN	12129	101582
6	PETROLEUM OIL	1478	44812
7	PROPAMOCARB HCL	0	44396
8	GLYPHOSATE	36880	40192
9	ETHOPROPHOS	0	34118
10	CHLOROTHALONIL	19081	19081
11	EPTC	18240	18580
12	PENDIMETHALIN	15734	17527
13	MANCOZEB	16354	16353
14	METRIBUZIN	15464	15539
15	PHOSPHORIC ACID	14143	14143
16	2,4-D	13242	13830
17	MCPA	12339	12441
18	DIMETHENAMID & DIMETHENAMID-P	7811	9466
19	DIMETHENAMID-P	7811	9200
20	MALEIC HYDRAZIDE	40	8254
21	SULFUR	919	8175
22	CHLORPYRIFOS	5882	6136
23	METOLACHLOR & METOLACHLOR-S	4726	5602
24	OXAMYL	5591	5591
25	METHOMYL	1	5187
26	METOLACHLOR-S	4568	5005
27	FLUROXYPYR	4698	4711
28	PROPARGITE	2	4692
29	BROMOXYNIL	4063	4329
30	DIQUAT	4271	4271
31	DIMETHOATE	542	4134
32	PARAQUAT	49	4122
33	PYRIMETHANIL	3356	3356
34	BOSCALID	3170	3171
35	IMIDACLOPRID	2805	2805
36	AZOXYSTROBIN	2701	2702
37	FLUTOLANIL	2462	2462
38	QUINTOZENE	240	2384
39	PYRACLOSTROBIN	1754	1879
40	SODIUM CHLORATE	1840	1840
41	NALED	0	1650
42	FLUOPYRAM	1326	1338
43	DIMETHENAMID	0	1329
44	PROPICONAZOLE	1086	1315
45	TERBACIL	0	1290
46	METIRAM	1010	1234
47	TERBUFOS	0	1232
48	COPPER HYDROXIDE	908	1210
49	CLOPYRALID	295	1181
50	GLUFOSINATE	1163	1163